Table 1: C3 per system, id (dev). BL = HyperKKL ($r = 4$; HKKL), BF = best model-based filter, BW = best WORKING model-based filter (below climatology; $k/n^{\dagger}$ = failed / present), DD = best data-driven same-data baseline. Changes: paired relative change of BL ± 95% CI half-width (Student-t over seeds; bold/underline: significant). FAILED filters are never used as the reference. Abbreviations: -or oracle, -Hu Huber, GB grey-box, Koopman-bi bilinear Koopman KF.

scenario / plant (clim.)	BL	NRMSE	BF	NRMSE	BW	NRMSE	fail.	(a) BL vs BW	DD	NRMSE	(b) BL vs DD
exact											
battery (1)	HyperKKL	0.0483	EKF	0.0059	EKF	0.0059	1/8	+718% ±71	pend. 0/5		–
bioreactor (1)	HyperKKL	0.0193	EKF-Hu	0.00622	EKF-Hu	0.00622	0/8	+211% ±46	pend. 0/5		–
duffing (1)	HyperKKL	0.0469	EKF	0.00461	EKF	0.00461	0/8	+916% ±90	pend. 0/5		–
lorenz (1)	HyperKKL	0.027	UKF	0.00659	UKF	0.00659	0/8	+310% ±101	pend. 0/5		–
quad_tank (1)	HyperKKL	0.0524	EKF	0.0144	EKF	0.0144	0/8	+264% ±17	pend. 0/5		–
vdp (1)	HyperKKL	0.0164	UKF	0.00561	UKF	0.00561	0/8	+192% ±5	pend. 0/5		–
exact+outliers											
battery (1)	HyperKKL	1.91	PF	0.0353	PF	0.0353	4/8	+5328% ±1402	pend. 0/5		–
bioreactor (1)	HyperKKL	0.117	EKF-Hu	0.00622	EKF-Hu	0.00622	1/8	+1790% ±858	pend. 0/5		–
duffing (1)	HyperKKL	2.27	UKF-Hu	0.00475	UKF-Hu	0.00475	1/8	+47710% ±6069	pend. 0/5		–
lorenz (1)	HyperKKL	0.607	UKF-Hu	0.0262	UKF-Hu	0.0262	2/8	+2215% ±57	pend. 0/5		–
quad_tank (1)	HyperKKL	1.38	EKF-or	0.0251	EKF-or	0.0251	3/8	+5380% ±231	pend. 0/5		–
vdp (1)	HyperKKL	0.854	EKF-Hu	0.00568	EKF-Hu	0.00568	2/8	+14939% ±740	pend. 0/5		–
colored_t											
battery (1)	HyperKKL	0.0476	EKF-or	0.0109	EKF-or	0.0109	1/8	+338% ±14	GB-EKF-or	0.0109	+337% ±20
bioreactor (1)	HyperKKL	0.0227	EKF-or	0.00625	EKF-or	0.00625	0/8	+263% ±189	GB-EKF-or	0.00626	+262% ±189
duffing (1)	HyperKKL	0.0694	EKF-or	0.00965	EKF-or	0.00965	0/8	+619% ±37	GB-EKF-or	0.00966	+619% ±37
lorenz (1)	HyperKKL	0.0441	UKF-Hu	0.0181	UKF-Hu	0.0181	0/8	+143% ±4	GB-EKF-or	0.0188	+134% ±3
quad_tank (1)	HyperKKL	0.0547	EKF-or	0.0172	EKF-or	0.0172	0/8	+218% ±34	GB-EKF-or	0.0173	+215% ±27
vdp (1)	HyperKKL	0.0314	EKF-or	0.0132	EKF-or	0.0132	0/8	+137% ±20	GB-EKF-or	0.0132	+137% ±20
mismatch10											
battery (0.743)	HyperKKL	0.0448	UKF	0.0574	UKF	0.0574	1/8	**-22%** ±17	GB-EKF	0.007	+540% ±164
bioreactor (0.794)	HyperKKL	0.02	EKF-or	0.372	EKF-or	0.372	5/8	**-95%** ±1	NLOX	0.0159	+28% ±38
duffing (0.995)	HyperKKL	0.048	EKF-or	0.0133	EKF-or	0.0133	0/8	+262% ±36	GB-EKF	0.00463	+937% ±104
lorenz (0.912)	HyperKKL	0.0215	UKF-or	0.158	UKF-or	0.158	0/8	**-86%** ±3	GB-EKF-or	0.0202	+7% ±33
quad_tank (0.554)	HyperKKL	0.0367	KKL-MB	0.429	KKL-MB	0.429	7/8	**-91%** ±2	GB-EKF	0.014	+162% ±28
vdp (1.06)	HyperKKL	0.0172	EKF-or	0.0244	EKF-or	0.0244	0/8	**-29%** ±8	GB-UKF	0.00601	+187% ±37
mismatch20											
battery (0.739)	HyperKKL	0.043	UKF-Hu	0.117	UKF-Hu	0.117	1/8	**-63%** ±5	GB-EKF	0.00977	+340% ±60
bioreactor (0.718)	HyperKKL	0.024	EKF-or	0.466	EKF-or	0.466	6/8	**-95%** ±2	NLOX	0.0149	+64% ±108
duffing (0.998)	HyperKKL	0.0458	EKF-or	0.0233	EKF-or	0.0233	0/8	+97% ±25	GB-EKF	0.00466	+884% ±131
lorenz (0.616)	HyperKKL	0.0069	KKL-MB	0.0146	KKL-MB	0.0146	0/8	**-53%** ±9	NLOX	0.00891	**-22%** ±15
quad_tank (0.534)	HyperKKL	0.0255	KKL-MB	$0.647^{\dagger}$	none	–	8/8	–	GB-EKF	0.0135	+89% ±30
vdp (1.07)	HyperKKL	0.0172	EKF-or	0.0388	EKF-or	0.0388	0/8	**-56%** ±5	GB-EKF	0.00706	+144% ±51
drift30 (post-train)											
battery (1)	HyperKKL	0.202	UKF-Hu	0.18	UKF-Hu	0.18	1/8	+12% ±2	pend. 0/5		–
bioreactor (1)	HyperKKL	4.63	EKF-or	0.512	EKF-or	0.512	6/8	+805% ±8	pend. 0/5		–
duffing (1)	HyperKKL	0.055	EKF-or	0.0355	EKF-or	0.0355	0/8	+55% ±10	pend. 0/5		–
lorenz (1)	HyperKKL	0.485	UKF-or	0.469	UKF-or	0.469	0/8	+4% ±1	pend. 0/5		–
quad_tank (1)	HyperKKL	1.57	KKL-MB	$1.5^{\dagger}$	none	–	8/8	–	pend. 0/5		–
vdp (1)	HyperKKL	0.0756	EKF-or	0.0701	EKF-or	0.0701	0/8	+8% ±1	pend. 0/5		–
mismatch30											
battery (0.735)	HyperKKL	0.0382	UKF-Hu	0.18	UKF-Hu	0.18	1/8	**-79%** ±4	GB-EKF	0.0127	+200% ±35
bioreactor (0.655)	HyperKKL	0.0329	EKF-or	0.505	EKF-or	0.505	6/8	**-93%** ±5	NLOX	0.0143	+140% ±251
duffing (1)	HyperKKL	0.0493	EKF-or	0.0329	EKF-or	0.0329	0/8	+50% ±32	GB-EKF	0.00477	+933% ±214
lorenz (0.614)	HyperKKL	0.00664	KKL-MB	0.0129	KKL-MB	0.0129	0/8	**-49%** ±14	NLOX	0.00669	-0% ±44
quad_tank (0.53)	HyperKKL	0.0221	KKL-MB	$0.711^{\dagger}$	none	–	8/8	–	GB-EKF	0.0134	+65% ±18
vdp (1.13)	HyperKKL	0.0179	EKF-or	0.0523	EKF-or	0.0523	0/8	**-66%** ±7	GB-EKF	0.00893	+101% ±35
mismatch50											
battery (0.736)	HyperKKL	0.0381	EKF	0.307	EKF	0.307	1/8	**-88%** ±5	GB-UKF	0.0185	+105% ±50
bioreactor (0.561)	HyperKKL	0.0501	UKF-or	$0.651^{\dagger}$	none	–	8/8	–	NLOX	0.0136	+285% ±397
duffing (1.01)	HyperKKL	0.0464	EKF-or	0.0529	EKF-or	0.0529	0/8	**-12%** ±11	GB-EKF	0.00526	+783% ±147
lorenz (0.409)	HyperKKL	0.00968	KKL-MB	0.0647	KKL-MB	0.0647	6/8	**-85%** ±17	NLOX	0.00386	+152% ±269
quad_tank (0.549)	HyperKKL	0.0246	KKL-MB	$0.962^{\dagger}$	none	–	8/8	–	GB-EKF	0.0133	+84% ±7
vdp (1.2)	HyperKKL	0.0185	KKL-MB	0.0637	KKL-MB	0.0637	0/8	**-71%** ±3	GB-EKF	0.0132	+40% ±30
partial											
battery (1)	HyperKKL	0.0476	UKF-or	0.397	UKF-or	0.397	1/8	**-88%** ±4	GB-UKF	0.027	+77% ±74
bioreactor (1)	HyperKKL	0.0239	UKF	$1.6^{\dagger}$	none	–	8/8	–	NLOX	0.017	+42% ±37
duffing (1)	HyperKKL	0.049	KKL-MB	0.101	KKL-MB	0.101	0/8	**-52%** ±15	GB-UKF	0.00732	+571% ±271
lorenz (1)	HyperKKL	0.0298	UKF-or	0.533	UKF-or	0.533	5/8	**-94%** ±3	NLOX	0.0285	+4% ±47
quad_tank (1)	HyperKKL	0.0517	EKF-or	0.0285	EKF-or	0.0285	0/8	+82% ±4	GB-EKF	0.015	+244% ±16
vdp (1)	HyperKKL	0.0157	KKL-MB	0.064	KKL-MB	0.064	0/8	**-75%** ±3	GB-EKF	0.0196	-18% ±36

Table 1: (continued)

scenario / plant (clim.)	BL	NRMSE	BF	NRMSE	BW	NRMSE	fail.	(a) BL vs BW	DD	NRMSE	(b) BL vs DD
realistic											
battery (0.735)	HyperKKL	0.0506	UKF-or	0.37	UKF-or	0.37	1/8	**-86%** ±1	GB-EKF-or	0.0185	+174% ±11
bioreactor (0.655)	HyperKKL	0.0313	KKL-MB	1.85†	none	–	8/8	–	NLOX	0.0171	+85% ±62
duffing (1)	HyperKKL	0.0697	UKF-or	0.143	UKF-or	0.143	0/8	**-51%** ±2	GB-EKF	0.0127	+448% ±24
lorenz (0.614)	HyperKKL	0.011	KKL-MB	0.0273	KKL-MB	0.0273	7/8	**-58%** ±10	NLOX	0.0134	**-18%** ±12
quad_tank (0.53)	HyperKKL	0.0245	KKL-MB	0.994†	none	–	8/8	–	GB-EKF	0.0168	+46% ±10
vdp (1.13)	HyperKKL	0.0294	KKL-MB	0.108	KKL-MB	0.108	0/8	**-73%** ±1	GB-EKF	0.031	-5% ±8

† FAILED: NRMSE above climatology (in brackets after the plant); fail. = FAILED / present model-based filters.